

Robert Orlando Lopez

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ABOUT ME

I've been working in software development for more than ten years, taking on projects that range from small internal tools to large-scale systems used by thousands of people. Over the years, I've learned how to balance writing clean, maintainable code with making sure features actually solve real problems for the business and end users. I enjoy working closely with product teams, mentoring other developers, and finding practical solutions when things get complicated. What motivates me most is seeing the impact of my work in production and knowing it makes life easier for both the company and its customers.

SKILLS

Programming Languages: Python (3.6–3.11), Java (8–11), C# (.NET Framework 4.0–4.6, .NET Core 3.1), SQL, JavaScript (ES6+), HTML5, CSS3

Frameworks & Platforms: Django (2.x–4.x), Django REST Framework, Flask, FastAPI, Celery, ASP.NET MVC, Java EE, Hibernate, AngularJS, React.js, jQuery

Databases: PostgreSQL (9–14), MySQL, Oracle Database, SQL Server, SQLite, Redis

APIs & Integration: REST APIs, GraphQL, JSON, XML, Swagger/OpenAPI

Cloud & DevOps: AWS (EC2, RDS, S3, Lambda, CloudWatch, SQS), Azure App Service, Docker, Jenkins, GitHub Actions, GitLab CI

Testing & Quality: pytest, unittest, Factory Boy, JUnit, MSTest, NUnit, Selenium

Monitoring & Observability: CloudWatch, Application Insights, ELK Stack, Prometheus, Grafana

Collaboration & Practices: Agile/Scrum, GitFlow, Code Reviews, CI/CD pipelines, SDLC best practices, Test-Driven Development

AI & LLM Integration (RAG / Copilots): Prompt engineering, structured JSON outputs, OpenAI / Azure OpenAI APIs, retrieval-augmented generation (RAG) pipelines, embedding-based retrieval workflows, LangChain-style orchestration patterns, evaluation/guardrail design for LLM calls

Vector Search & Data: pgvector, Pinecone, Azure AI Search, vector-based similarity search, schema design for obligations/controls/policies/evidence, change-log/versioning models

Document Processing & Ingestion: PDF/HTML parsing, text normalization and chunking, OCR tooling, Azure Document Intelligence-style pipelines, regulation ingestion workflows (download, parse, version, diff)

Cloud (Azure Focus): Azure Functions (serverless pipelines), Azure Blob Storage, Azure SQL, basic Azure DevOps CI/CD, containerized microservices running on Azure-backed infrastructure

EXPERIENCE

Apex Systems – Glen Allen, VA

Senior Full Stack Engineer | Programming Analyst

Apr 2019 – Present

- Designed and developed high-performance microservices using Python 3.8+ with FastAPI framework, leveraging Pydantic for comprehensive data validation and type-safe request/response handling
- Built scalable APIs with async/await patterns using Uvicorn and Gunicorn ASGI servers, implementing asynchronous I/O operations that reduced response latency by 60% for real-time features
- Optimized database operations using SQLAlchemy ORM with PostgreSQL, implementing Alembic migrations and Redis caching layers to improve query performance and reduce database load

- Implemented robust background job processing with Celery and Redis message broker, handling long-running tasks including report generation, data synchronization, and third-party API integrations
- Containerized applications using Docker and deployed to AWS ECS with auto-scaling policies, implementing comprehensive CI/CD pipelines for automated testing, security scanning, and rolling deployments
- Architected cloud-native solutions leveraging AWS services including S3 for object storage, RDS for managed databases, Lambda for serverless functions, and CloudWatch for comprehensive monitoring
- Enhanced system reliability through comprehensive test coverage using pytest and pytest-asyncio, ensuring robust testing of asynchronous routes, database operations, and API integrations
- Designed and implemented an LLM-driven extraction pipeline to pull obligations and key clauses from PDF/HTML policy and regulatory documents, normalizing outputs into structured JSON compatible with existing FastAPI services
- Built an end-to-end RAG workflow: ingestion (download, parse, chunk), embedding generation, vector storage (pgvector / Pinecone), and a retrieval layer orchestrated through FastAPI endpoints to power internal “regulatory copilot”-style APIs
- Implemented change detection jobs that periodically re-ingest source regulations, diff new versions against prior snapshots, and surface impact analysis (e.g., which obligations, controls, and systems are affected) via asynchronous Celery tasks
- Modeled a mapping engine linking obligations → internal controls → policies → evidence, using PostgreSQL schemas designed for auditability, version history, and traceability from LLM outputs back to source documents
- Collaborated with frontend development teams using React.js with modern hooks and context APIs, designing efficient data exchange patterns and optimizing API performance for responsive user interfaces
- Implemented secure authentication and authorization flows using OAuth2/OpenID Connect standards, integrating with enterprise identity providers and implementing role-based access control
- Integrated Azure Functions as serverless workers for document ingestion and pre-processing, coordinating with Azure Blob Storage for raw and processed documents and Azure SQL/Postgres for metadata and obligation storage
- Prototyped LLM endpoints using Azure OpenAI and OpenAI APIs with prompt templates focused on deterministic, schema-constrained JSON outputs, including validation and fallbacks when responses did not conform to the expected structure
- Exposed REST APIs that behave as a “Regulatory Copilot” backend: streaming answer endpoints that combine retrieved context from vector search with LLM reasoning, and metadata APIs that surface underlying citations and confidence scores
- Set up CI/CD automation for the AI/RAG pipelines, including unit tests for chunking/embedding logic, contract tests for LLM prompts, and deployment workflows for containerized microservices running in cloud environments
- Established advanced monitoring and observability using Prometheus metrics collection, Grafana dashboards, and distributed tracing to maintain optimal system performance and rapid incident response
- Mentored development teams on modern Python practices, asynchronous programming patterns, and cloud-native architecture principles through code reviews and technical guidance sessions

UEAT – Quebec, Canada

Senior Software Engineer

Aug 2017 – Mar 2019

- Developed and maintained scalable backend services using Python 3.6+ with Flask framework, implementing Flask-RESTful for clean API design and Flask-SQLAlchemy for efficient database operations
- Designed and optimized database schemas using Flask-Migrate for version control, implementing efficient transactional queries for PostgreSQL and MySQL environments supporting high-volume order processing

- Built comprehensive authentication and session management systems using Flask-Login with OAuth2 integration, supporting multiple third-party authentication providers for seamless user experience
- Architected modular application structure using Flask blueprints to enable microservices scalability, refactoring legacy monolithic components into maintainable, reusable service modules
- Implemented asynchronous task processing using Celery with Redis message broker, enabling background operations for email dispatch, report generation, and real-time order tracking functionality
- Deployed containerized services using Docker with basic orchestration on AWS EC2 instances, implementing load balancing and auto-scaling to handle variable traffic loads efficiently
- Integrated AWS cloud services including S3 for static asset storage, RDS for managed database hosting, and CloudWatch for application monitoring and alerting capabilities
- Enhanced development workflows by establishing comprehensive pytest testing suites and integrating automated test execution into early CI pipeline implementations
- Experimented with early recommendation and ranking services using Python data pipelines and PostgreSQL, laying groundwork for future AI-powered personalization features
- Built internal analytics endpoints that exposed normalized, queryable order data for downstream reporting tools and potential ML model consumption
- Collaborated with frontend development teams to design and implement API integrations, ensuring optimal data flow and responsive user experiences for the restaurant ordering platform
- Contributed to system architecture decisions and technology stack modernization, helping transition from legacy systems to more scalable and maintainable solutions

Canva – Austin, TX

Full Stack Engineer

Aug 2013 – Mar 2017

- Developed backend services using Java with Spring Framework, creating RESTful APIs that supported collaborative design workflows, user management, and media processing features
- Implemented efficient data persistence using Hibernate ORM with PostgreSQL, designing optimized database schemas and query strategies to handle large-scale user-generated content and collaboration data
- Built responsive frontend interfaces using AngularJS and jQuery, creating seamless integration with backend APIs to enable real-time content editing and interactive user dashboards
- Contributed to system architecture improvements by refactoring monolithic components into modular services, implementing clean separation of concerns and improving code maintainability
- Introduced containerization technologies starting with Docker adoption in 2015, improving development environment consistency and laying groundwork for more scalable deployment strategies
- Deployed and scaled applications on AWS infrastructure using EC2 for compute resources, S3 for media asset storage, and RDS for managed relational database hosting
- Established automated testing practices using JUnit for backend services and basic JavaScript testing frameworks, improving code reliability and supporting more frequent release cycles
- Collaborated closely with product managers and UX designers to translate creative workflow requirements into robust technical implementations while maintaining system performance at scale
- Participated in technology evaluation and adoption decisions, contributing to the selection of tools and frameworks that supported rapid feature development and platform growth
- Documented API specifications and integration patterns, facilitating effective collaboration between frontend and backend development teams and improving developer onboarding processes

Ciena – Hanover, MD

Intern | Junior Developer

Jun 2012 – Jun 2013

- Contributed to enterprise application development using C# with .NET Framework 4.x, focusing on data access layers and business logic implementation for telecommunications operations and reporting systems
- Implemented database integration modules using ADO.NET for Oracle Database and SQL Server connectivity, writing stored procedures, parameterized queries, and transaction management logic
- Developed internal tools and reporting interfaces using ASP.NET MVC 4, while maintaining and enhancing existing ASP.NET Web Forms applications for legacy system support
- Created business intelligence dashboards using Crystal Reports integrated with .NET applications, providing analytics and operational insights for telecommunications management teams
- Enhanced user interfaces by implementing JavaScript and jQuery functionality, improving the usability and interactivity of internal tools and employee-facing applications
- Participated in enterprise deployment processes to VMware-based infrastructure, learning configuration management and build validation procedures under senior developer supervision
- Gained experience with version control systems including SVN and Git migration processes, learning collaborative development workflows and branching strategies in enterprise environments
- Implemented comprehensive unit testing using NUnit framework, learning test-driven development principles and contributing to regression testing and quality assurance processes

EDUCATION

Illinois Institute of Technology

May 2012

Bachelor of Computer Science; Cumulative GPA: 3.5/4.0